

Cambridge International AS & A Level

MATHEMATICS**9709/32**

Paper 3 Pure Mathematics 3

May/June 2026**1 hour 50 minutes**

You must answer on the answer booklet.

You will need: Answer booklet
List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Write your answers on the answer booklet. Follow the instructions on the front cover of the answer booklet. If you need additional answer paper, ask the invigilator for a continuation booklet.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **4** pages.

- 1 Find the quotient and remainder when $4x^3 - 4x^2 + 7x + 2$ is divided by $2x - 1$. [3]

- 2 Solve the equation $2^{5-2x} = \frac{4^{x+1}}{3}$. Give your answer in the form $\frac{\ln a}{\ln b}$, where a and b are integers. [4]

- 3 The complex numbers s and t are given by

$$s = 3(\cos 2 + i \sin 2) \quad \text{and} \quad t = 2e^{-i}.$$

- (a) Express st in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$ and $r > 0$. [2]

- (b) In an Argand diagram with origin O , the points A and B represent the complex numbers s and st respectively.

By considering the line segments OA and OB , or otherwise, state the two geometric effects on OA of multiplying s by t . [2]

- 4 The square roots of $3 - (6\sqrt{2})i$ can be expressed in the Cartesian form $x + iy$, where x and y are real and exact.

By first forming a quartic equation, find the square roots of $3 - (6\sqrt{2})i$. Give your answer in exact Cartesian form. [5]

- 5 The parametric equations of a curve are

$$x = 2\theta - \tan 2\theta, \quad y = 6\theta + 2 \ln(\cos 2\theta),$$

for $0 \leq \theta \leq \frac{1}{4}\pi$.

- (a) Find an expression for $\frac{dy}{dx}$ in terms of $\tan 2\theta$. [6]

- (b) Find the exact coordinates of the point on the curve at which the gradient of the tangent is equal to -1 . [4]

- 6 The equation of a curve is $y = x \tan^{-1} 3x$ for $-\pi \leq x \leq \pi$.

- (a) Find the exact value of the gradient of the curve when $x = \frac{1}{3}$. [4]

- (b) Find the exact value of $\int_0^{\frac{1}{3}} y \, dx$. [6]

- 7 The equation of a curve is

$$y = \frac{3 + \sin 3x}{2 - \cos 3x}$$

for $0 \leq x \leq \pi$. The gradient of the curve is equal to $-\frac{3}{2}$ when $x = a$.

- (a) Show that a satisfies the equation $\sin^2 3a + 6 \sin 3a - 3 = 0$. [4]

- (b) Hence find the values of a for $0 \leq a \leq \pi$. [3]

- 8 (a) By sketching a suitable pair of graphs, show that the equation $\operatorname{cosec} \frac{1}{2}x = \frac{1}{2}(x+3)$ has exactly one root in the interval $0 \leq x \leq \pi$. [2]

- (b) Show by calculation that this root lies between 1 and 1.2. [2]

- (c) Show that, if a sequence of real values given by the iterative formula

$$x_{n+1} = 2 \sin^{-1} \left(\frac{2}{x_n + 3} \right)$$

converges, then it converges to the root of the equation in 8(a). [2]

- (d) Use the iterative formula in 8(c) to calculate this root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

- 9 The points A and B have position vectors $\begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix}$ respectively.

The line m passes through the points A and B .

- (a) Find a vector equation for m . [2]

The line l has equation $\mathbf{r} = \begin{pmatrix} 6 \\ -1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$.

- (b) Determine whether the lines l and m are parallel, intersect or are skew. [4]

- (c) The point P lies on l and is such that angle $PAB = 60^\circ$.

Find the position vector of P . [6]

Question 10 is on the back page.

10 (a) Express $\frac{x+3}{(x-1)(2x-6)}$ in partial fractions. [4]

(b) The variables x and y satisfy the differential equation

$$(x-1)(2x-6)\frac{dy}{dx} = \frac{x+3}{\ln(2y)}.$$

It is given that $y = 1$ when $x = 4$.

Solve the differential equation to obtain an equation in x and y . [7]

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